

TEMASEK JUNIOR COLLEGE

2023 JC2 PRELIMINARY EXAMINATION

Higher 2



CANDIDATE
NAME

ANSWER

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GROUP

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CENTRE
NUMBER

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INDEX
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Chemistry

9729/04

Paper 4 Practical

28 August 2023

2 hours 30 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your centre number, index number, name and CG on all the work you hand in.
Give details of the practical shift and laboratory where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.
Qualitative Analysis Notes are printed on pages 19 and 20.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
4	
Total	/ 55

This document consists of 18 printed pages and 2 blank page.

Answer **all** the questions in the spaces provided.

- 1 In this experiment you are to determine the relative formula mass of an iron(II) salt by titration with potassium manganate(VII).

FA 1 is the iron (II) salt.

FA 2 is $0.0100 \text{ mol dm}^{-3}$ potassium manganate(VII), KMnO_4 .

FA 3 is 1.0 mol dm^{-3} sulfuric acid, H_2SO_4 .

Method

(a) Preparing a solution of FA 1

1. Weigh accurately about 4.0 g of **FA 1** into the plastic vial labelled **FA 1** and record the mass in the space below.
2. Transfer all the weighed **FA 1** into a 250 cm^3 beaker.
3. Use a measuring cylinder to add approximately 100 cm^3 of **FA 3** to the beaker. Stir until all the solid has dissolved.
4. Transfer the solution into the 250 cm^3 volumetric (graduated) flask labelled **FA 4**.
5. Wash out the beaker thoroughly using distilled water and add the washings to the volumetric flask. Make the solution up to the mark using distilled water.
6. Shake the flask thoroughly to mix the solution before using it for your titrations. Label this solution of the iron(II) salt as **FA 4**.

Mass of plastic vial/g	6.01
Mass of plastic vial and FA 1/g	9.91
Mass of plastic vial and residue/g	6.01
Mass of FA 1 used/g	3.90

[2]

[1] Table with correct headings and units

[1] All readings to 2 or 3 d.p

(b) Titration of FA 4 against FA 2

1. Fill the burette with **FA 2**.
Retain the burette containing FA 2 for Q2.
2. Pipette 25.0 cm^3 of **FA 4** into a conical flask.
3. Use a measuring cylinder to add 20 cm^3 of **FA 3** to the flask.
4. Titrate **FA 4** with **FA 2** until the solution changes to a permanent pink colour.
5. Record your titration results, to an appropriate level of precision, in the space provided below.
6. Repeat steps 2 to 5 until consistent results are obtained.

(i) Titration results

Titration No.	1	2
Final burette reading / cm ³	22.10	43.45
Initial burette reading / cm ³	0.70	22.10
Volume FA 2 used / cm ³	21.40	21.35
	✓	✓

[1] Table with correct headings and units

[1] All readings to 2 d.p.

[2]

- (ii) From your accurate titration results, obtain a suitable value to be used in your calculations. Show clearly how you have obtained this value. [4]

Average volume of FA 2 used = $(21.40 + 21.35)/2 \text{ cm}^3 = 21.38 \text{ cm}^3$

[1] Show correct working using consistent readings

Accuracy:

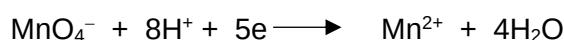
[3m] supervisor reading $\pm 0.03 \text{ cm}^3 \text{ g}^{-1}$ [2m] supervisor reading $\pm 0.06 \text{ cm}^3 \text{ g}^{-1}$ [1m] supervisor reading $\leq 0.10 \text{ cm}^3 \text{ g}^{-1}$

(c) Calculations

- (i) Calculate the number of moles of potassium manganate(VII) present in the volume of FA 2 calculated in (b)(ii). [1]

Amount of $\text{MnO}_4^- = 21.38/1000 \times 0.01 \text{ mol} = 2.14 \times 10^{-4} \text{ mol}$ [1]

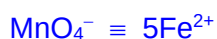
- (ii) The half-equation for the reduction of a manganate(VII) ion is:



Write the complete balanced ionic equation for the oxidation of iron(II) ion to iron(III) by manganate(VII) ions. [1]



- (iii) Calculate the number of moles of iron(II) ions present in 25.0 cm³ of solution FA 4. [1]



Amount of Fe^{2+} in 25.0 cm³ = $2.14 \times 10^{-4} \times 5 \text{ mol} = 1.07 \times 10^{-3} \text{ mol}$ [1]

- (iv) Calculate the number of moles of iron(II) ions present in 250 cm³ of solution FA 4. [1]

$$\text{Amount of Fe}^{2+} \text{ in } 250 \text{ cm}^3 = 1.07 \times 10^{-3} \times 10 \text{ mol} = 1.07 \times 10^{-2} \text{ mol} \quad [1]$$

- (v) Calculate the relative formula mass of the iron(II) salt. [3]

$$\text{Amount of salt} = 1.07 \times 10^{-2} \text{ mol}$$

$$\text{Mass of salt weighed} = 3.90 \text{ g}$$

$$M_r = 3.90 / 1.07 \times 10^{-2} = 364.5 \quad [1]$$

[1] Shows working in all calculations. All calculations must be relevant although they may not be complete or correct. Any calculation not attempted loses this mark.

[1] Shows appropriate significant figures (3 or 4 sf) and units in all final answers. Any calculation not attempted loses this mark.

- (d) A student conducted the above experiments and obtained an average titre value of 19.30 cm³.

The errors (uncertainties) associated with **each reading** using a graduated flask, pipette and burette are $\pm 0.15 \text{ cm}^3$, $\pm 0.10 \text{ cm}^3$ and $\pm 0.05 \text{ cm}^3$ respectively.

Calculate the maximum total percentage error (uncertainty) of this mean titre volume. [1]

$$\text{Maximum \% error} = \pm (0.15/250 + 0.10/25.0 + 2 \times 0.05/19.30) \times 100$$

$$= \pm (6 \times 10^{-4} + 4 \times 10^{-3} + 5.18 \times 10^{-3}) \times 100$$

$$= \pm 0.978 \quad [1]$$

[Total: 16]

- 2 Glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, is a sugar that can act as a reducing agent. You will investigate how an increase in temperature affects the rate of the redox reaction between glucose and acidified potassium manganate(VII).

FA 2 is $0.010 \text{ mol dm}^{-3}$ potassium manganate(VII), KMnO_4 .

FA 3 is 1.0 mol dm^{-3} sulfuric acid, H_2SO_4 .

FA 5 is an aqueous solution containing 32.8 g dm^{-3} glucose, $\text{C}_6\text{H}_{12}\text{O}_6$.

distilled water

You will measure the time it takes for the purple colour to disappear. Your table of results on page 8 should include the rate of reaction for each experiment.

(a) Method

Experiment 1

1. Add 10.00 cm^3 of **FA 2** from the burette (in question 1) into the 250 cm^3 beaker.
2. Use the 100 cm^3 measuring cylinder to transfer 50.0 cm^3 of **FA 3** into the beaker containing **FA 2**.
3. Use the same measuring cylinder to transfer 50.0 cm^3 of distilled water into the same beaker.
4. Place the beaker on the tripod and heat its contents to between 65°C and 70°C .
5. While the solution in the beaker is heating pour 25.0 cm^3 of **FA 5** into the 25 cm^3 measuring cylinder.
6. When the temperature of the contents of the beaker has reached between 65°C and 70°C , turn off the Bunsen burner and **carefully** place the hot beaker onto the white tile.
7. Record the initial temperature, T_i , of the solution in the beaker.
8. Add the 25.0 cm^3 of **FA 5** and **immediately** start timing.
9. Stir the contents of the beaker once and stop timing as soon as the solution turns colourless. Record the time to the nearest second.
10. Record the final temperature, T_f , of the solution as soon as it is colourless.
11. Calculate and record the average temperature, T_{avg} , of the reaction mixture to one decimal place.
12. Repeat Experiment 1 at four different temperatures, between room temperature and 70°C , each time using a clean and dry beaker. Record all your results in a table on page 8.

Results

The rate of reaction can be calculated as shown.

$$\text{Rate} = \frac{1000}{\text{reaction time}}$$

Calculate the rate of reaction to 3 sf for each experiment and include this in your table.
[4]

Expt	Initial Temp / °C	Final Temp / °C	Average Temp / °C	Time / s	Rate / s ⁻¹
1	69.0	67.0	68.0	11	90.9
2	59.0	54.0	56.5	26	38.5
3	49.0	45.5	47.3	55	18.2
4	40.0	38.0	39.0	112	8.93
5	29.5	30.0	29.8	335	2.99

[1] Table to show 5 experiment data with 5 headings + units

[1] Temp readings recorded to .0 or .5 **and** all times as integers. All average temperatures correctly calculated to 1 d.p.

[1] All rates correctly calculated to 3 s.f. All rates increase with increase temperature.

[1] Selected initial temperatures in Experiment 2 to 5 that are at least 5 °C apart and none above 70 °C.

- (b) Plot a graph of rate (y-axis) against average temperature (x-axis) on the grid opposite. Select a scale on the x-axis to include an average temperature of 15.0 °C. Label any points you consider anomalous.

Draw a line of best fit and extrapolate it to 15.0 °C. [4]

[1] Axes correctly labelled and scale chosen so graph occupies more than half the available length for both axes including 15 °C on the x-axis

Points in 6 large squares on y-axis and y-axis must not go below 0

[1] All points from data (min 4 expt) accurately plotted within ½ small square (any point that is supposed to be on the line must be on the line and any point that is supposed to be inside a small square must not be on the boundary line.

[1] Line of best curve

[1] Curve extrapolated to 15 °C

- (c) **Use your graph** to calculate the **time** to the nearest second that the reaction would have taken if the average temperature had been 52.5 °C. Show **on the grid** how you obtained your answer. [2]

[1] Correct reading of rate within ½ small square and two construction lines shown

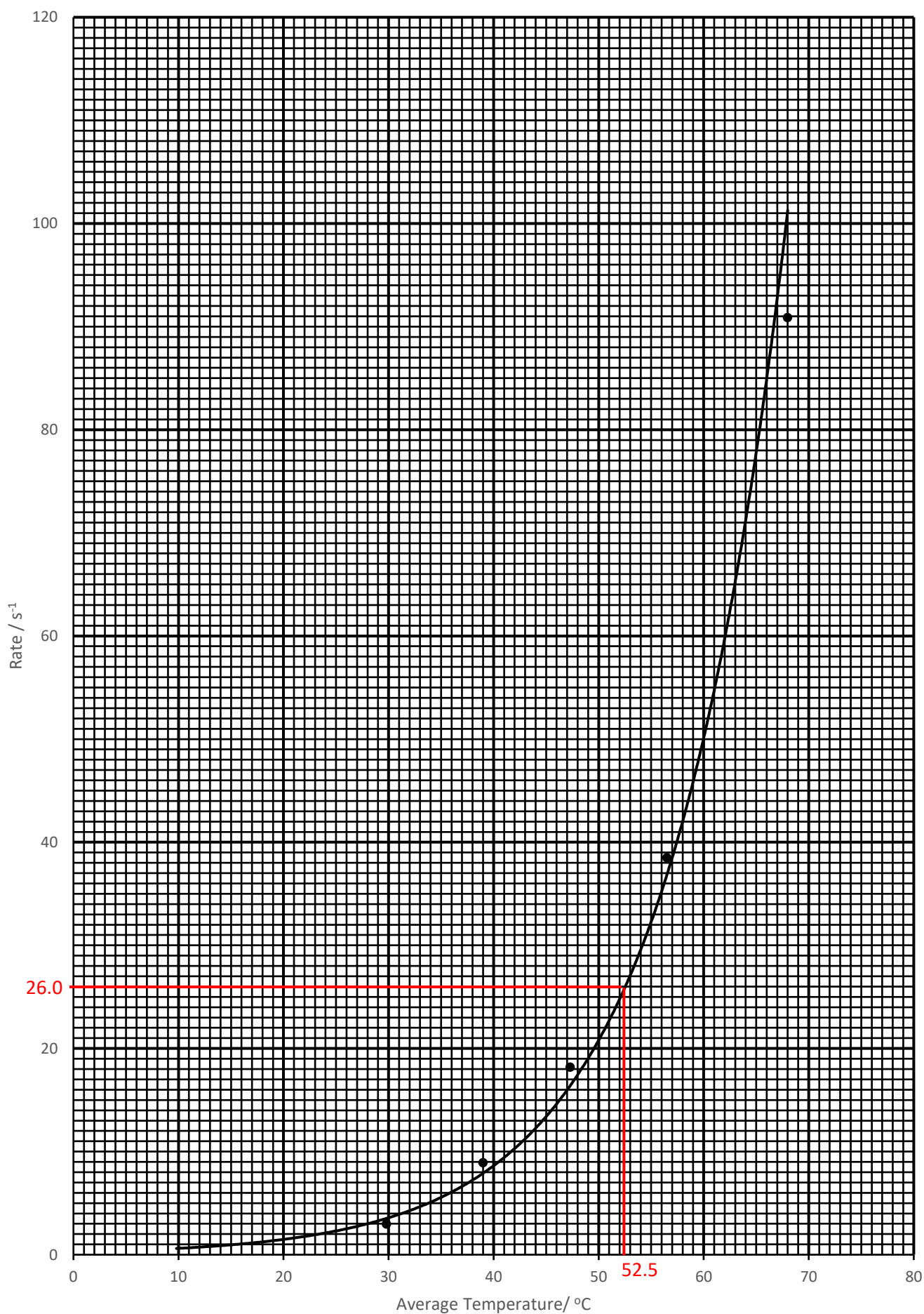
When average temperature = 52.5 °C, from the graph, rate = 26.0

[1] $\text{Time} = \frac{1000}{\text{rate}} = \frac{1000}{26} = 38 \text{ s}$

- (d) Explain, by referring to your graph or your table of results, how the rate of reaction is affected by an increase in temperature. [1]

[1] The rate of reaction increases with temperature or rate is proportional to temperature.

[1]



- (e) Calculate the maximum percentage error in the reaction time recorded for **Experiment 1**. Assume the error of the timer is $\pm 1\text{s}$. [1]

[1] Maximum percentage error = $\pm \frac{1}{\text{time}} \times 100 = \pm \frac{1}{11} \times 100 = \pm 9.09\%$

- (f) Suggest **two** ways to improve the accuracy of the results for this investigation. [2]

Any 2 of the following

- Use thermostatically controlled water bath (to heat both reagents / keep reagents at const T).
- Use more precisely calibrated thermometer (allow more precise but not more accurate or more sensitive) not 'use a digital thermometer'
- Use light sensor / colorimeter (to avoid subjective judgement of colour fade).
- Use burette / measuring cylinders calibrated to greater precision (to measure volumes of FA3 and FA5)

[Total: 14]

3 Qualitative Analysis

At each stage of any test you are to record details of the following.

- colour changes seen
- the formation of any precipitate
- the solubility of such precipitates in an excess of the reagent added

Where gases are released they should be identified by a test, **describe at an appropriate juncture of your observations.**

You should indicate clearly at what stage in a test a change occurs.

No additional tests for ions present should be attempted.

If any solution is warmed, a boiling tube MUST be used.

Rinse and reuse test-tubes and boiling tubes where possible.

Where reagents are selected for use in a test, the name or correct formula of the element or compound must be given.

(a) FA 6 and **FA 7** are aqueous solutions of equal concentrations, in mol dm^{-3} . Each contains one anion and one cation. The cation is the same in both **FA 6** and **FA 7**.

Half-fill the 250 cm^3 beaker with water. Heat the water to about 80°C and then **turn off the Bunsen burner**. This is the hot water bath needed for the tests below.

To about 2 cm depth of aqueous silver nitrate in a test-tube, add a few drops of aqueous sodium hydroxide to give a grey / brown precipitate. Then add aqueous ammonia dropwise until the precipitate **just** disappears. This solution is Tollens' reagent and is needed in a test below.

(i) Carry out the tests on separate samples of **FA 6** and **FA 7** and complete the table.

Precaution : FA 6 should be kept stoppered at all times.

test	observations	
	FA 6	FA 7
To a 1 cm depth of solution in a test-tube in a test-tube rack, add a spatula measure of sodium carbonate.	✓ Effervescence. ✓ Colourless, acidic gas evolved. Gives white ppt. with Ca(OH)_2.	✓ Effervescence. ✓ Colourless, acidic gas evolved. Gives white ppt. with Ca(OH)_2.
To a 1 cm depth of solution in a test-tube, add a few drops of acidified potassium manganate(VII). Place the test-tube in the hot water bath.	✓ Purple solution ✓ Solution turns colourless / decolourise	✓ Purple solution ✓ No decolourisation / remains purple
To a 1 cm depth of Tollens' reagent in a test-tube, add a few drops of solution. Place the test-tube in the hot water bath and leave for several minutes.	✓ Grey/black ppt or silver mirror obtained	(White ppt accepted) ✓ No grey/black ppt or silver mirror obtained

9-10 ✓ **3m**; 5-8 ✓ **2 m**; 2-4 ✓ **1m**

[3]

(ii) From your observations in (i), identify the cation present in **both FA 6 and FA 7**.

[1]

cation H^+ [1]

(iii) From your observations in (i), what can be deduced about the anion present in **FA 6**? [1]

[1] Can be oxidized / reducing agent/ contains aldehyde group

(iv) Place a 1 cm depth of **FA 6** and **FA 7** separately in two test-tubes.

Measure and record the temperature of the two solutions.

FA 6 ✓28.0..... °C

FA 7 ✓28.0..... °C

To each solution, add an approximately 2 cm length of magnesium ribbon. Measure and record the maximum temperature reached in each test-tube.

FA 6 + Mg ✓36.0..... °C

FA 7 + Mg ✓40.0..... °C

[2]

2✓ **1m** (Temperature with Mg must be higher than that without Mg)

(v) Explain why there is a difference in the temperature rise for the reactions of magnesium with solutions **FA 6** and **FA 7**.

[1] FA 6 is a weak/weaker acid. The [1] heat released from the reaction of the acid with Mg is used to further dissociate the acid and hence the temperature measured is lower than that of FA 7.

[2]

(b) **FA 8** is a solution containing a mixture of two cations from those listed on page 19.

You are to plan a series of experiments that will enable you to identify the cations present. You should then carry out your plan, record all the observations you made in a suitable table and identify the cations present.

Test	Observation
✓To 1 cm depth of FA 8, add NaOH(aq) dropwise until in excess.	✓ <u>White ppt. soluble in excess NaOH</u> to give a colourless solution.
✓Warm the mixture.	✓Colourless, pungent, <u>alkaline gas evolved. Turned damp red litmus blue.</u>
✓To 1 cm depth of FA 8, add NH ₃ (aq) dropwise until in excess.	✓ <u>White ppt. soluble in excess NH₃(aq)</u> to give a colourless solution.

3✓ : 1m

Cations present are[1]Zn²⁺.....and.....[1] ...NH₄⁺.....

[5]

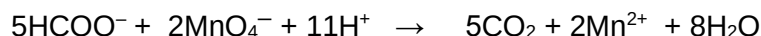
[Total: 14]

4 Planning

Magnesium methanoate, $\text{Mg}(\text{HCOO})_2$ has been used in studies to develop a multiresidue method for determining pesticides in fruits and vegetables.

A saturated solution of magnesium methanoate has a solubility of approximately 150 g dm^{-3} at room temperature. 50 cm^3 of the saturated solution was diluted to 250 cm^3 . This diluted solution is **FB 1**.

You are to conduct an experiment to determine the exact solubility of the saturated solution of magnesium methanoate by titrating against acidified potassium manganate (VII) as shown below. [M_r of $\text{Mg}(\text{HCOO})_2 = 114.3$]



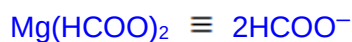
You may assume you are provided with:

- 250 cm^3 of **FB1**, the diluted magnesium methanoate, $\text{Mg}(\text{HCOO})_2$, solution
- 0.02 mol dm^{-3} aqueous potassium manganate (VII), KMnO_4
- 100 cm^3 of 1.0 mol dm^{-3} sulfuric acid, H_2SO_4
- 250 cm^3 volumetric flask
- the equipment normally found in a school or college laboratory

- (a) Verify that the approximate concentration of HCOO^- in **FB1** is $0.525 \text{ mol dm}^{-3}$. [1]

$$\text{Saturated } [\text{Mg}(\text{HCOO})_2] = \frac{150}{114.3} = 1.312 \text{ mol dm}^{-3}$$

$$\text{Diluted } [\text{Mg}(\text{HCOO})_2] = 1.312 \times \frac{50}{250} = 0.2624 \text{ mol dm}^{-3}$$



$$\text{Diluted } [\text{HCOO}^-] = 2 \times 0.2624 = \underline{0.525 \text{ mol dm}^{-3}}$$

[1] for correct working i.e. $\div$ by 114.3, $\times$ 2 mole ratio and dilute 5 times.

- (b) The concentration of HCOO^- in part (a) is unsuitable for titration with KMnO_4 .

Dilution is required in the volumetric procedure in order to obtain a reliable titre value.

Calculate a suitable volume of **FB 1** to be further diluted taking into account the equipment found in the college laboratory. [2]



Method 1 : Calculate [HCOO^-]

Assuming a titre volume of 25 cm^3 of MnO_4^- ,

$$\text{Amount of } \text{MnO}_4^- = 25/1000 \times 0.02 = 5.00 \times 10^{-4} \text{ mol}$$



$$\text{Amount of } \text{HCOO}^- = 5/2 \times 5.00 \times 10^{-4} = 1.25 \times 10^{-3} \text{ mol}$$

Assume using a 25 cm^3 pipette,

$$[\text{HCOO}^-] = n/v = \frac{0.00125}{0.025} = \underline{0.0500 \text{ mol dm}^{-3}} \quad [1]$$

$$\text{Dilution factor} = \frac{0.525}{0.0500} = 10.5$$

Using a 250 cm³ volumetric flask for dilution,

Volume of Mg(HCOO)₂ to be diluted = 250/10.5 = 23.80 cm³ [or round up to whole number].
[1] for answer

Method 2: Calculate volume of MnO₄⁻ needed
Assume using 25 cm³ pipette,

$$\text{Amount of HCOO}^- \text{ in } 25 \text{ cm}^3 = \frac{25.00}{1000} \times 0.5248 = 0.01313 \text{ mol}$$

$$\text{Amount of MnO}_4^- = 2/5 \times 0.01313 = 0.00525 \text{ mol}$$

$$\text{Volume of MnO}_4^- = c/v = \frac{0.00525}{0.02} = \underline{0.262 \text{ dm}^3 \text{ or } 262 \text{ cm}^3} \quad [1]$$

This volume of MnO₄⁻ in the burette exceeds the burette capacity hence is not suitable.

Hence, there is a need to dilute the given concentration of Mg(HCOO)₂.

Assuming a titre value of 25 cm³ is suitable [accept any range between 20-30 cm³],

$$\text{Dilution factor} = \frac{262}{25} = 10.5$$

Using a 250 cm³ volumetric flask for dilution,

Volume of Mg(HCOO)₂ to be diluted = 250/10.5 = 23.80 cm³ [or round up to whole number].
[1] for answer

- (c) Plan an experiment to determine the exact solubility of the saturated solution of magnesium methanoate, taking into account your answers in (b).

Your plan should include details of:

- then apparatus you would use
- the procedure you would follow
- the measurements you would take

[4]

Dilution

1. Using a burette, transfer 23.80 cm³ of filtrate (range over 23.80 cm³) into a 250 cm³ volumetric flask
2. Top up to the mark with distilled water and shake to obtain a homogenous solution
3. Label this solution FA1 / diluted Mg(HCOO)₂

Titration

1. Pipette 25.0 cm³ of diluted Mg(HCOO)₂/ FA1 into a conical flask
2. Using a measuring cylinder, add 25 cm³ of dilute H₂SO₄ (an excess) into the conical flask.
3. Fill a burette with KMnO₄.
4. Titrate the diluted Mg(HCOO)₂/ FA1 with KMnO₄ until 1 drop of the titre turns the mixture from colourless to pale/permanent pink. Record your titre value.
5. Repeat the titration for consistent results (i.e. results with a difference of ± 0.10 cm³ or less).

D: [1] For dilution procedure

P: [2] For titration procedure (including addition of H₂SO₄, colour change and repeat until consistent results) ;

A: [1] for choice of apparatus and state quantities (If Dilution procedure not written, mark awarded for correct titration apparatus)

- (d) A student diluted x cm³ of FB1 to 250 cm³ and obtained y cm³ of KMnO₄ for the titration.

- (i) Calculate the concentration of HCOO⁻ in FB1. [2]

Let the average titre value of MnO₄⁻ = y cm³

Amount of MnO₄⁻ = $0.020 \times y/1000 = 2 \times 10^{-5} y$

[1] Amount of HCOO⁻ in 25 cm³ = $5/2 \times (2 \times 10^{-5} y) = 5 \times 10^{-5}$

Amount of HCOO⁻ in 250 cm³ = No mol of HCOO⁻ in x cm³ (volume diluted)

= $250/25 \times 5 \times 10^{-5} y = 5.0 \times 10^{-4} y$

[1] $[HCOO^-] = \frac{0.0005y}{x/1000} = \frac{0.5y}{x} \text{ mol dm}^{-3}$

- (ii) Hence, calculate the concentration of HCOO⁻ in saturated solution of magnesium methanoate. [1]

[1] $[HCOO^-] = \frac{0.0005y}{x/1000} \times 5 = \frac{2.5y}{x} \text{ mol dm}^{-3}$

- (iii) Determine the solubility product of magnesium methanoate. [1]

$[Mg^{2+}] = \frac{1}{2} [HCOO^-] = \frac{1}{2} \times \left(\frac{0.0005y}{x/1000} \right) \times 5 = \frac{1.25y}{x} \text{ mol dm}^{-3}$

[1] $K_{sp} = [Mg^{2+}] [HCOO^-]^2$

= $\left(\frac{1.25y}{x} \right) \left(\frac{2.5y}{x} \right)^2 \text{ mol}^3 \text{ dm}^{-9}$

= $\frac{125 y^3}{16 x^3} \text{ mol}^3 \text{ dm}^{-9}$ **[with correct units]**

[Total: 11]

Qualitative Analysis Notes

[ppt. = precipitate]

(a) Reactions of aqueous cations

<i>cation</i>	<i>reaction with</i>	
	NaOH(aq)	NH ₃ (aq)
aluminium, Al ³⁺ (aq)	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, NH ₄ ⁺ (aq)	ammonia produced on heating	–
barium, Ba ²⁺ (aq)	no ppt. (if reagents are pure)	no ppt.
calcium, Ca ²⁺ (aq)	white ppt. with high [Ca ²⁺ (aq)]	no ppt.
chromium(III), Cr ³⁺ (aq)	grey-green ppt. soluble in excess giving dark green solution	grey-green ppt. insoluble in excess
copper(II), Cu ²⁺ (aq)	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), Fe ²⁺ (aq)	green ppt., turning brown on contact with air insoluble in excess	green ppt., turning brown on contact with air insoluble in excess
iron(III), Fe ³⁺ (aq)	red-brown ppt. insoluble in excess	red-brown ppt. insoluble in excess
magnesium, Mg ²⁺ (aq)	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), Mn ²⁺ (aq)	off-white ppt., rapidly turning brown on contact with air insoluble in excess	off-white ppt. rapidly turning brown on contact with air insoluble in excess
zinc, Zn ²⁺ (aq)	white ppt. soluble in excess	white ppt. soluble in excess

DO NOT WRITE IN THIS MARGIN

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(b) Reactions of aqueous anions

<i>anion</i>	<i>reaction</i>
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, $\text{Cl}^-(\text{aq})$	gives white ppt. with $\text{Ag}^+(\text{aq})$ (soluble in $\text{NH}_3(\text{aq})$)
bromide, $\text{Br}^-(\text{aq})$	gives pale cream ppt. with $\text{Ag}^+(\text{aq})$ (partially soluble in $\text{NH}_3(\text{aq})$)
iodide, $\text{I}^-(\text{aq})$	gives yellow ppt. with $\text{Ag}^+(\text{aq})$ (insoluble in $\text{NH}_3(\text{aq})$)
nitrate, $\text{NO}_3^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil
nitrite, $\text{NO}_2^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil; NO liberated by dilute acids (colourless $\text{NO} \rightarrow$ (pale) brown NO_2 in air)
sulfate, $\text{SO}_4^{2-}(\text{aq})$	gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (insoluble in excess dilute strong acids)
sulfite, $\text{SO}_3^{2-}(\text{aq})$	SO_2 liberated with dilute acids; gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (soluble in dilute strong acids)

(c) Test for gases

<i>gas</i>	<i>test and test result</i>
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	"pops" with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns aqueous acidified potassium manganate(VII) from purple to colourless

(d) Colour of halogens

<i>halogen</i>	<i>colour of element</i>	<i>colour in aqueous solution</i>	<i>colour in hexane</i>
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas / liquid	orange	orange-red
iodine, I_2	black solid / purple gas	brown	purple